

Md. Abu Sayed Chowdhury

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EDUCATION

Bangladesh University of Engineering and Technology (BUET) 2025–Present

M.Sc. in Electrical and Electronic Engineering, in progress

Focus: VLSI design and emerging computing technologies

Bangladesh University of Engineering and Technology (BUET) 2020–2025

B.Sc. in Electrical and Electronic Engineering

CGPA: 3.95/4.00; Class rank: 7/213. Dean's List in all academic years.

Relevant coursework: Advanced VLSI Design; VLSI Circuits and Design; Analog Integrated Circuits; Semiconductor and Nano Devices; Nanoelectronics; Processing and Fabrication Technology; Machine Learning and Pattern Recognition; Spintronics.

RESEARCH INTERESTS

In-memory computing with memristor/ReRAM crossbars; hybrid memristor–CMOS architectures; neuromorphic and energy-efficient AI hardware; low-power VLSI and mixed-signal circuit design.

PUBLICATIONS

M. A. S. Chowdhury and O. Hassan, “In-Memory Boolean Logic Computation in Binary Neural Network Inspired Memristor Crossbar Array,” in *2026 10th IEEE Electron Devices Technology & Manufacturing Conference (EDTM)*, 2026.

DOI: 10.1109/EDTM65772.2026.11497945 | IEEE Xplore

RESEARCH EXPERIENCE

Undergraduate Thesis Research, BUET Jan 2024–Mar 2025

In-Memory Logic Implementation in Memristor-Based Crossbar Arrays

Supervisor: Dr. Orchi Hassan

- Developed a binary-neural-network-inspired approach to in-memory Boolean logic, encoding each binary weight with two memristor devices in a crossbar array.
- Designed and simulated CMOS programming and readout peripherals in Cadence Virtuoso using GPDK045; demonstrated NAND and XOR gates and cascaded full-adder operation through circuit-level SPICE simulations.
- Implemented and calibrated VTEAM-based Verilog-A compact models to represent threshold switching and non-linear current–voltage behavior for CMOS–memristor co-simulation.
- First author of the resulting conference paper published in IEEE EDTM 2026 proceedings.

ACADEMIC APPOINTMENTS & TEACHING

Lecturer, BRAC University Jan 2026–Present

Department of Electrical and Electronic Engineering, BSRM School of Engineering

- Teaching areas: Signals and Systems (EEE243), Numerical Techniques (EEE282), and Modeling and Simulation (EEE382).

Adjunct Lecturer, BUET Apr 2025–Jan 2026

Department of Electrical and Electronic Engineering

- Conducted laboratory sessions in Electrical Circuits, Numerical Techniques, and Digital Signal Processing, including pre-lab briefings, demonstrations, and experimental troubleshooting.
- Contributed to laboratory curriculum and assessment updates; provided structured feedback and mentoring to support students' technical and conceptual development.

SELECTED TECHNICAL PROJECTS

16-bit Carry Skip Adder: RTL-to-GDSII Design

Dec 2024

SystemVerilog, Cadence Genus/Innovus, GPDK045 | GitHub

- Implemented block-based carry-skip RTL and a SystemVerilog verification environment with constrained-random stimulus and functional coverage; achieved 100% coverage of eight targeted signed-arithmetic flag scenarios.
- Compared block sizes $k = 2, 4, 8$ through synthesis and power, performance, and area analysis; selected $k = 4$ with a 1.24 ns critical-path delay, 3.88 μW total power, and 136.8 μm^2 cell area.
- Completed floorplanning, power distribution, placement, clock-tree synthesis, and routing in Innovus; exported GDSII with clean DRC/LVS and no setup or hold violations.

Analog 8:1 Multiplexer for Biomedical Signal Acquisition

Feb 2024

Cadence Virtuoso, GPDK045, CMOS transmission gates | GitHub

- Designed an eight-channel transmission-gate multiplexer for ECG/EEG signal acquisition; optimized transistor dimensions to reduce crosstalk and signal distortion.
- Characterized transient, AC, DC, and noise behavior: 71.8 μW power, 1.576 pC charge injection, 92.66 ns switching delay, 3.5 MHz -3 dB bandwidth, and approximately 850 Ω on-resistance at a 3 V supply.

FPGA-Based Intelligent Home Automation

Aug 2023

Verilog, Intel Quartus, Altera Flex EPF10K70 | GitHub

- Implemented finite-state machines for bidirectional occupancy counting, occupancy-triggered lighting with a 10-second shutoff, and four-speed PWM fan control (25–100%).
- Integrated infrared sensors and NTC thermistor modules to activate the fan above 30 $^{\circ}\text{C}$ and adjust speed according to occupancy.

TECHNICAL SKILLS

- **IC design and EDA:** Cadence Virtuoso (schematic/layout), Genus (synthesis), Innovus (place-and-route, clock-tree synthesis); GPDK045; DRC/LVS verification.
- **Hardware description and modeling:** Verilog, SystemVerilog, Verilog-A; VTEAM memristor modeling; RTL verification, functional coverage, finite-state machines, and FPGA development with Intel Quartus Prime.
- **Simulation and programming:** SPICE-based circuit simulation, LTspice, PSpice, Proteus; Python (NumPy, pandas, PyTorch), MATLAB/Simulink, C/C++, ARM assembly.
- **Embedded platforms:** STM32, Arduino, ESP8266/ESP32, and Altera FPGAs.

HONORS & AWARDS

- **First Runner-Up (2nd Place), VLSITHON 2024:** University-wide VLSI design competition organized by Neural Semiconductor in collaboration with BUET EEE Day 2024.
- **University Merit Scholarship, BUET:** Awarded for five semesters for academic excellence.